



June 2025

climate risk in vulnerable mountain regions of the world

Day 3 (29 June)

Jakob Steiner

Course content

27 June (1400 – 1800)

Pre course evaluation - general introduction – the mathematics of risk – introduction to assignments

28 June (1000 – 1800)

Examples of hazards / exposure types /vulnerability assessments

Communicating risk

Science to policy

29 June (1000 – 1800)

Serious game – Course wrap up

Course content

29 June (1000 – 1800)

10:00 – 10:45	International Platforms to address Mountain Issues
10:45 - 11:00	Break
11:00 - 13:00	Intro to the Serious Game – preparations and National Deliberations
13:00 - 14:00	Lunch Break
14:00 - 15:00	COP Deliberations
15:00 - 15:15	Break
15:15 - 16:15	Plenary Session
16:15 - 17:00	Recap
17:00 - 18:00	Course Feedback, Evaluation

Course content

29 June (1000 – 1800)

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International initiatives addressing mountain risks

Loss and Damage (<https://www.lossanddamagecollaboration.org>)

Mountain Research Initiative (MRI) (<https://mountainresearchinitiative.org/>)

Adaptation at Altitude (<https://adaptationataltitude.org/>)

Mountain Partnership (<https://www.fao.org/mountain-partnership/home/en/>)



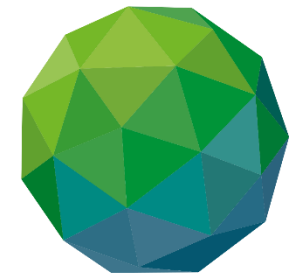
United Nations
Framework Convention on
Climate Change

UNFCCC and Loss and Damage

United Nations Framework Convention on Climate Change (UNFCCC)



- An international treaty to limit dangerous climate change
- Signed by 154 states in 192 (Rio de Janeiro)
- Secretariat in Bonn, Germany
- Hosts the Green Climate Fund (GCF) – goal of \$100 billion annually by 2020 – the Adaptation Fund (AF), as well as smaller LDCF, SCCF and GEF
- Invites the IPCC (Intergovernmental Panel on Climate Change) to provide assessments
- Conference of Parties (COPs) are arranged under the framework (many of them!)



**GREEN
CLIMATE
FUND**



United Nations
Convention to Combat
Desertification

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UNCCD 16th session of the Conference of the Parties

RIYADH, SAUDI ARABIA | 2-13 DECEMBER 2024

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[Special and side events](#)

[Media](#)

Loss and Damage Fund

[...] Parties recognised by the UNFCCC as the [most responsible](#) for the climate crisis would provide predictable, and new and additional finance to address loss and damage.

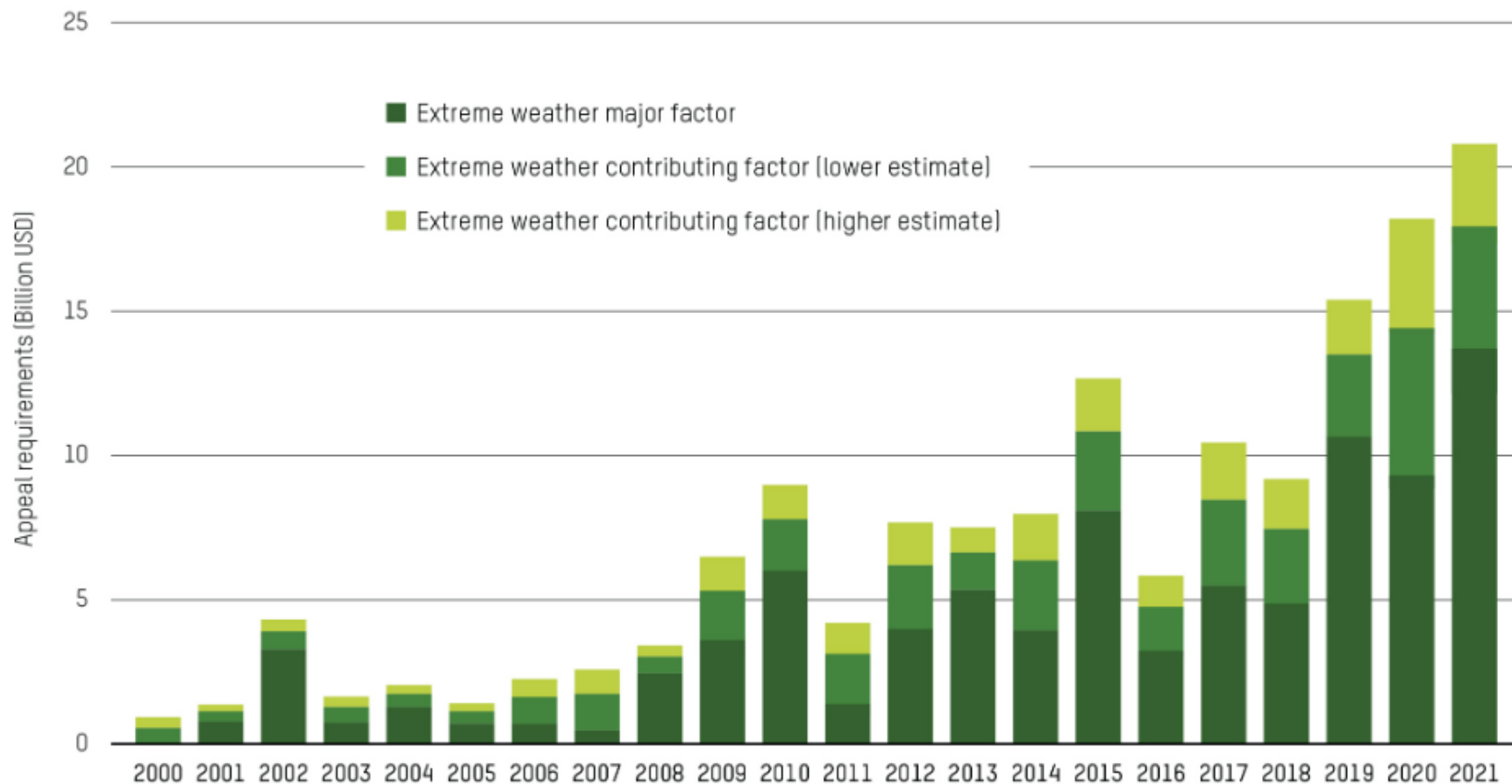
[...] Loss and Damage Fund will tackle the gaps that current climate finance institutions such as the [Green Climate Fund](#) (GCF) and humanitarian assistance [...] do not fill on Loss and Damage. The current international climate finance system is already [stretched to breaking point](#) due to [insufficient funds](#), lack of political will and poor quality climate finance that indebts countries.

Loss and Damage Fund

[...] Parties recognised by
provide predictable, and r

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[point](#) due to [insufficient fur](#)
indebts countries.

Figure 3: Funding requirements for UN humanitarian appeals linked to extreme weather, 2000–2021



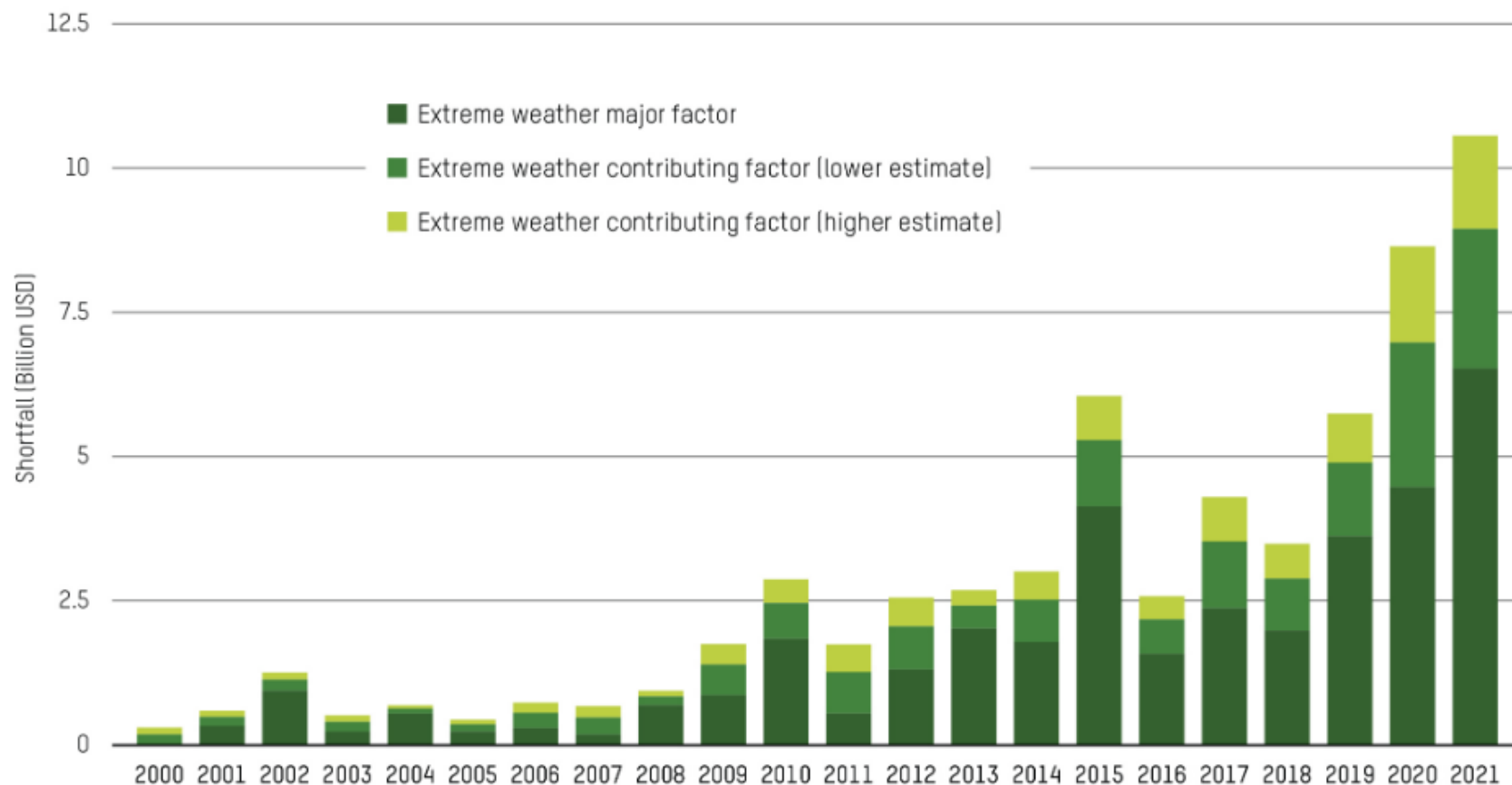
USD amounts are in 2021 real terms. The dark green sections cover appeals for which extreme weather was a major factor, and therefore 100% of the appeal requirement is counted. The higher estimate (lightest green) counts 50% and the lower estimate (middle green) counts 30% of the value of these appeals.

Loss and Damage Fund

[...] Parties recognised by the Fund to provide predictable, and not

[...] Loss and Damage Fund as the [Green Climate Fund](#) Damage. The current international [point](#) due to [insufficient funding](#) indebts countries.

Figure 4: Funding shortfalls for extreme-weather-related UN appeals, 2000–2021



Notes: USD amounts are in 2021 real terms. The dark green sections cover appeals for which extreme weather was a major factor, and therefore 100% of the appeal shortfall is counted. The higher estimate (lightest green) counts 50% and the lower estimate (middle green) counts 30% of the value of the appeal shortfall.

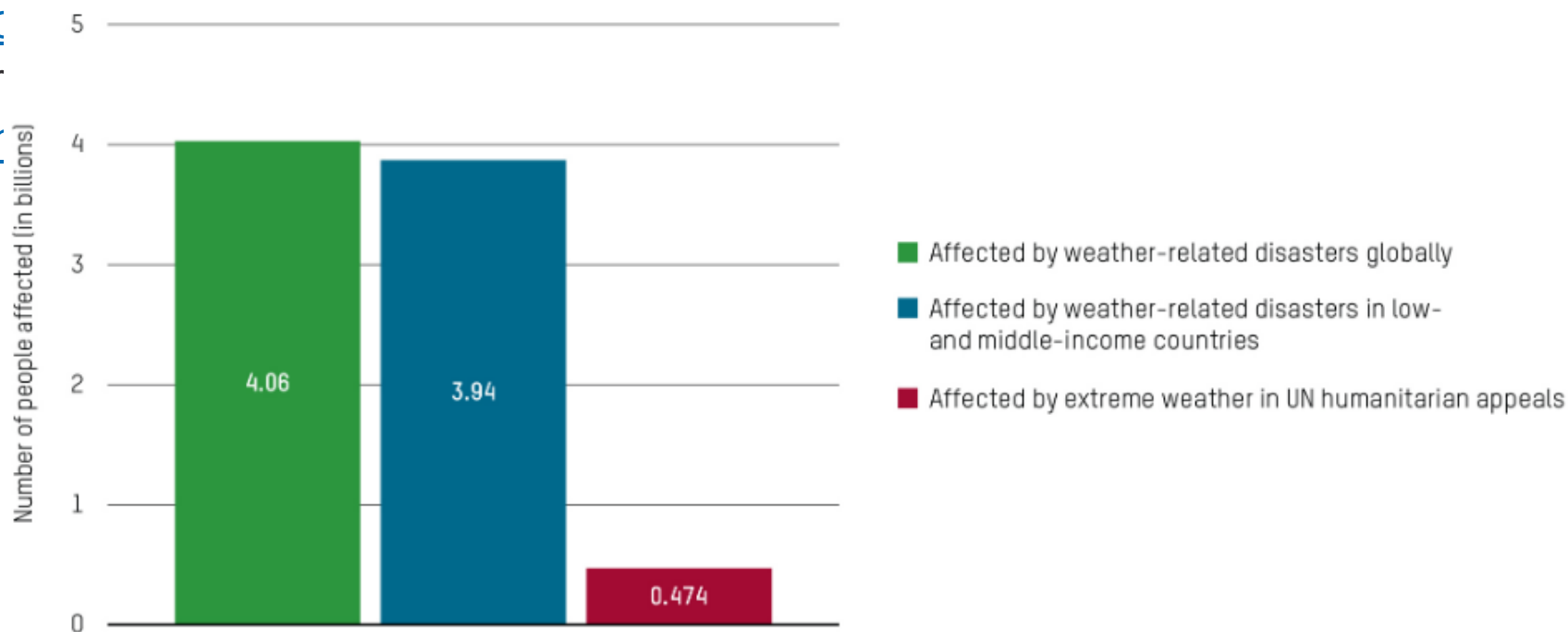
Loss and Damage Fund

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inadequacy of data
for mountain regions
hrisk.org

Figure 6: Number of people affected by weather-related disasters, 2000–2021



Source: Oxfam calculations using EM-DAT database.⁴⁴

Loss and Damage

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h risk.org

Table 3: Estimated future costs of loss and damage

Source, year	Estimated costs
<i>IFRC 'The cost of doing nothing' (2019)⁵⁹</i>	2030: humanitarian funding requirements for climate-related disasters could grow to \$20bn per year.
<i>Economist Intelligence Unit (2019)⁶⁰</i>	Measuring each country's direct exposure to loss from climate change and extreme weather events found that by 2050, GDP at risk could be: Africa (most vulnerable) 4.7% Latin America 3.8% Middle East 3.7% Eastern Europe 3% Asia Pacific 2.6% Western Europe 1.7% North America 1.1%
<i>Markandya and González-Eguino (2018)⁶¹</i>	Total annual residual damages for developing countries: 2030: \$290–\$580bn 2040: \$551bn–\$1tn 2050: \$1.1–\$1.7tn
<i>Climate Analytics (2015)⁶²</i>	Projected macro-economic damage of climate change for developing countries: 2030: \$400–\$431bn 2050: \$1–\$1.8tn
<i>Deloitte (2022)⁶³</i>	3°C trajectory could cost the U.S. economy \$14.5 trillion by 2070.

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Where do mountains come in?

Continuous attempts to bring *mountains* on the agenda since 1992

Less congruence in approach than SIDS (Small Island Developing States)

In some regions (like High Mountain Asia) complicated by political conflict

Often central governments themselves not located in mountains (perceived *marginal issue*) – leading to an increased emphasis of ‘*upstream-downstream linkages*’ and ‘*cascading risks*’

Mountain Partnership (FAO)

Formal and government led

MOUNTAIN PARTNERSHIP STEERING COMMITTEE



The Steering Committee of the Mountain Partnership provides programmatic orientation to the Mountain Partnership, oversees the preparation of a four-year strategy and a two-year workplan. The 18-member body monitors the work of the Mountain Partnership Secretariat, the implementation of the 'Mountain Agenda' and the achievements and impacts of their respective electoral groups.

Steering Committee members are elected to four-year terms by the various membership categories and regions. They meet annually and select a Chair and Vice Chairs.

The following governments and organizations are serving on the 2022-2025 Steering Committee:

Donor Organization to the MPS Italy (Chair)

Governments

Asia and the Pacific: Nepal

Europe: Andorra, Romania (Alternate)

Central Asia: Kyrgyzstan

Near East and North Africa: Yemen

North and Central America and the Caribbean: Dominican Republic (Vice Chair), Guatemala (Alternate)

South America: Peru, Argentina (Alternate)

Sub-Saharan Africa: Malawi, Uganda (Alternate)

Host Institution of the MPS

Food and Agriculture Organization of the United Nations

Intergovernmental Organization

United Nations Environment Programme, United Nations University (Alternate)

- > Our Vision and Mission
- > Governance and Strategy
- > **Mountain Partnership Steering Committee**
- > Mountain Partnership Secretariat
- > Logos and PowerPoint presentations
- > FAQ
- > Definitions

2024

- > MP Steering Committee meeting report (22 February 2024)

2023

- > MP Steering Committee meeting report (26 October 2023)
- > MP Steering Committee meeting report (20 April 2023)

2022

- > MP Steering Committee meeting report (1 December 2022)
- > MP Steering Committee meeting report (29 September 2022)
- > MP Steering Committee meeting report (30 August 2022)
- > MP Steering Committee meeting report (21 June 2022)
- > MP Steering Committee meeting report (30 March 2022)
- > MP Steering Committee Documents (13 January 2022)

2021

- > MP Steering Committee

Mountain Range Organisations

Alps: Alpine Convention - <https://www.alpconv.org/en/>

- Convention that includes a lawyer and proposes legal texts that are not binding but are used by arbitration courts in case of inter-governmental as well as country-internal disagreements

Carpathians: Carpathian Convention <http://www.carpathianconvention.org/>

- Similar to above, less developed/younger

Andes: CONDESAN <https://condesan.org/>

- Network of organisations working in the Andes (research heavy)

Hindukush-Himalaya: ICIMOD <https://www.icimod.org/>

- Intergovernmental organization (8 countries) that supports governments with expertise and provides platform for exchange

Mountain Research Initiative

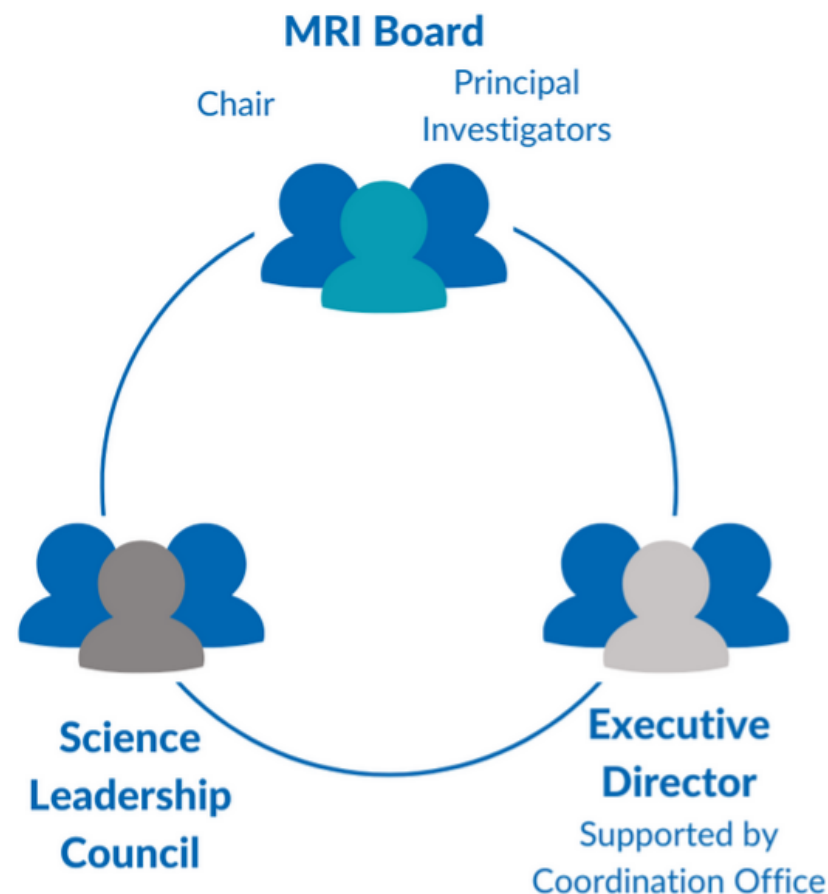
MRI Governance

The MRI Board (Chair and co-Principal Investigators), together with the MRI Science Leadership Council and the MRI Executive Director, constitute the MRI Governing Body. Together, they oversee and provide strategic direction and advice to support the implementation and development of both flagship and community-led activities, with the support of the Coordination Office staff.

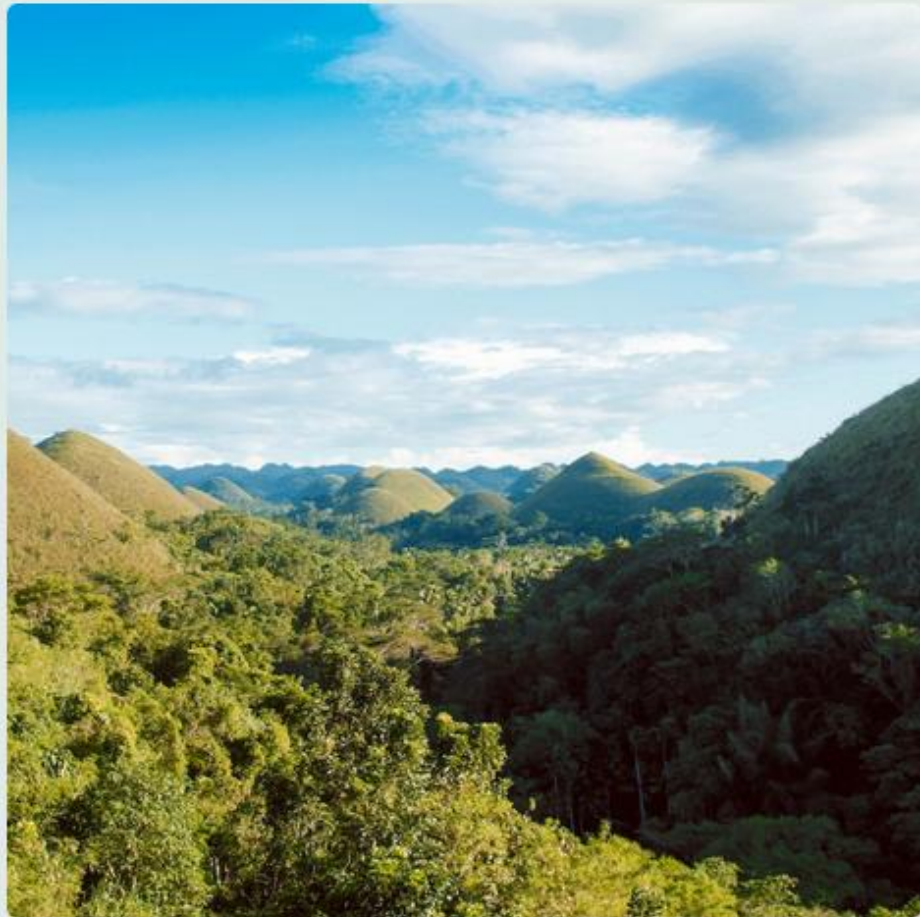
As a team, we focus on supporting the mountain research community in making meaningful and fulfilling connections for our changing mountains and safeguarding them as social-ecological treasures for ecosystems and people worldwide.

Learn more about us:

- [Principal Investigators](#)
- [Science Leadership Council](#)
- [Coordination Office](#)



Adaptation at Altitude



Our Objective

The Adaptation at Altitude programme seeks to increase the resilience and adaptive capacity of mountain communities and ecosystems to climate change by:

- Improving the knowledge of appropriate climate change adaptation strategies in the mountains
- Transferring that knowledge through science-policy platforms to inform decision-making in national, regional and global policy processes

Adaptation at Altitude will foster exchange among the mountain areas of the world and seek short and long-term solutions to the problems arising from climate change.

International Mountain Conference



✉ support@imc2025.info



SUMMERSCHOOL

International Mountain Conference

September 14 - 18 2025

CONFERENCE CORNER ▾

VENUES ▾

TRAVEL INFO ▾

CONTACT ▾

ABOUT ▾

#IMC DASHBOARD ▾



Mountains of Knowledge



Knowledge of Mountains

#imc25

Who is missing?

EXTREME HITZE

Hitze in Europa: Lufttemperatur am Mont Blanc über Gefrierpunkt

Auf Sardinien hat es aktuell bis zu 41 Grad. Menschen verlassen in Massen die italienischen Städte, auf den Autobahnen herrscht starker Verkehr

29. Juni 2025, 10:40

Auch das Meerwasser ist außergewöhnlich warm: 27 Grad an der ligurischen Küste und im zentralen Tyrrhenischen Meer, stellenweise sogar 28 bis 29 Grad im südlichen Tyrrhenischen Raum, so die Meteorologen.

Klimaanlagen, Eis und Stadtflucht

Die Italiener investieren zunehmend in Klimaanlage, um der Hitze zu entkommen – die Nachfrage ist stark gestiegen. Auch der Verbrauch von Speiseeis nimmt deutlich zu. Diese starke Nachfrage hat die Preise in vielen Eisdielen ansteigen lassen: Laut Konsumentenschutzverbänden sind die Preise für Eis in Italien in den letzten drei Jahren um 30 Prozent gestiegen.

Bereits am Samstag hatte eine Massenflucht aus den Städten eingesetzt. Am Sonntagmorgen wurde äußerst starker Verkehr auf den Autobahnen gemeldet. Destinationen der Hitzegeplagten waren das Meer und kühlere Bergregionen. (APA, 29.6.2025)



UN Climate Change Conference – Belém, November 2025

10 Nov – 21 Nov 2025

133

DAYS

21

HOURS

50

MINUTES

16

SECONDS



Previous

Next



COP Deliberations

Find common ground in focus groups:

Academic

International Organisations and ministers

Advocacy and Civil Society

Plenary Session – Country Strategies

Bhutan: knowledge of farmers, integrating with technological improvements and scientific research (drones, dual benefit); increase input from communities in the mountains; hydropower dependence, adaptation strategies as glacier ice reduces -> other renewable energy resources (wind), require financing;

Monitor forest cover, carbon sequestration (carbon negative country)

Glacial lake outburst flood monitoring – is less significant directly for society, lack of common ground

Approach India during deliberations – export/import dependence

Plenary Session – Country Strategies

New Zealand: integration of indigenous knowledge to risk assessment, include the wider society – developed a model to do risk assessment, focus on ecosystem, issue of invasive species, need data from other countries -> create transboundary initiative to exchange knowledge and data with Indigenous knowledge – see potential with other nations. Strengthen ecotourism, would be less dependent on other countries; benefit Indigenous communities; other countries can learn from that experience

Plenary Session – Country Strategies

Tanzania: water resources challenges, serious water scarcity, large population; water quality/pollution (industrial waste/agricultural use) leading to health issues/impacts, affecting food security – need to invest in education, foster competitions for innovation at University level. Exchange of talent with other countries (academic exchange). Strengthen governance/laws around water pollution (include NGOs/private sector to address together)

Treat mountain adaptation as a peace and development issue (connection between droughts e.g. and civil issues)

Plenary Session – Strategies from Interest Groups

International Organizations and ministers:

Differences in priorities (short term issues widely different), common ground when it comes to protection of species and ecosystems, water supply – including Indigenous People in strategy development. Climate Risk Assessment and Adaptation knowledge needs to be shared in a transboundary and transparent manner (remote sensing!) – exchange with academics

Work together when approaching other countries that have more financial capacities to address issues and take responsibility.

Plenary Session – Strategies from Interest Groups

Academics:

Transboundary initiative for data collection (incl Indigenous knowledge) – exchange monitoring data

Include local populations in collecting data (citizen science!) – academics analyse

Individual assessment of priorities across countries

How to make this profitable? Who would be willing to give money? – example India for Bhutan; Ecotourism (restrict tourism, locals guide)

Plenary Session – Strategies from Interest Groups

Advocacy and Civil Society:

Agreed on building an online network for exchanging data, knowledge – need help from data experts, track progress; integrate Indigenous voices; document the exchanges/videos

Work with and collaborate with countries from existing import/export partnerships

Plenary Session - Agreements

- Including Indigenous knowledge and local society into Science/data collection and bring that on the table to discuss
- Expand data collection centrally (e.g. through remote sensing) to have a basis for (transparent) discussion
- Establishing a global fund focus on transboundary/ecosystem issues – not enough players present, we could offer them something
- Difficult agreement on timelines (e.g. 5 years to have current data)
- Ecotourism wasn't picked up as an immediate goal

COP is officially over – you're asked by your media to report success or failure

Did you achieve what you wanted to get through? Where did you need to compromise?

Is there self criticism for your own country, your own interest group?

Did you see potential to find global common ground for interests of mountain regions?

How would you change your strategy for next time?

What stakeholder need to be brought in for next deliberations?

COP is officially over – you're back to your Self

Feedback to you:

You independently found argument lines in a very short time!

You were able to approach each other very quickly in uncomfortable territory on two levels (other students you do not know, countries/backgrounds that are not your own)!

You addressed many of the real topics that are being discussed in the real life scenario without me having assigned them up front.

You tried to cross boundaries!

COP is officially over – you're back to your Self

Could you find agreements with others? What made coming to agreements or strategies most difficult? What were you lacking to come to agreements?

Is virtual exchange useful? Do you need to see the others in person, do you need to travel to their countries?

Do you think it is realistic to come to agreements in such a setting and what would it take that they are being adhered to?

Feedback